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# Project Presentation:

## Digital Guitar Effect Pedal Prototype with Audio DSP

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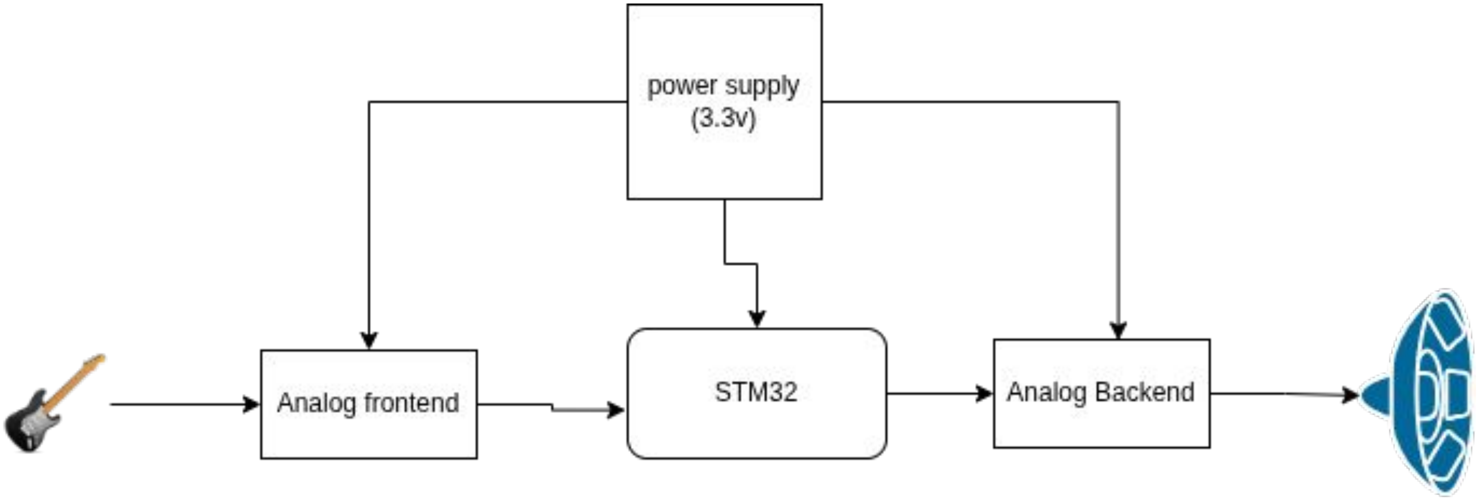
# Introduction

## What is it? What does it do? Why build it?

- **Concept:** Guitar effect pedals take sound from guitar input apply effects/distortions to it and forward to the guitar amp
- **What was done different:** Traditionally guitar pedals are done with analog components. Digital Effect pedals allow more complex effect and to be modified easily without changing hardware
- **How was it done?** By using DSP concepts we can digitally process audio with a microcontroller. The way the audio signal is transferred to the digital word is through the use of ADC and DAC components. It is practical application of many theoretical concepts learned at school.
- **Outcome:** Portable functional effect box that implements “Echo” effects to audio.

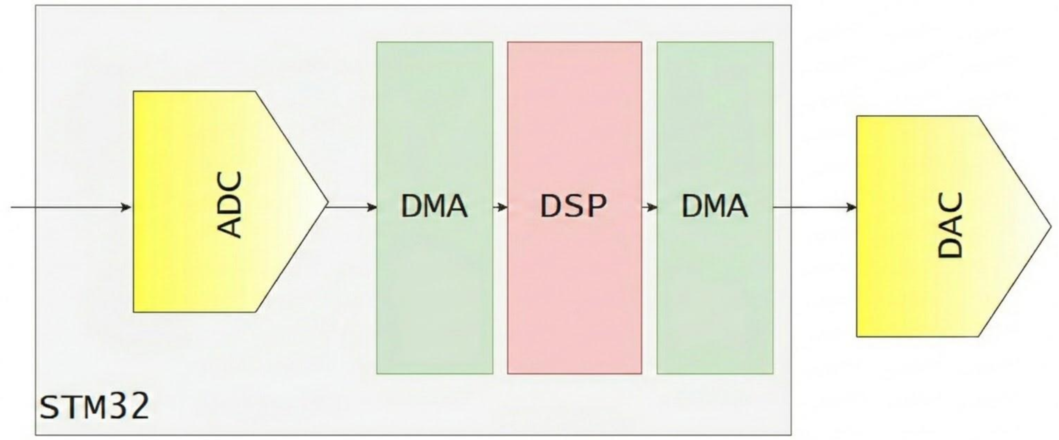
# System Components Overview




- **Analog Frontend:** Makes guitar signal fit ADC input range. Sets impedance, adds 1.65 V DC bias, buffers with op-amp, filters at 15.9 kHz.
- **STM32H743 (Cortex-M7, 480 MHz):** Samples at 48 kHz with hardware timer + DMA. Runs delay algorithm while data transfers happen in background.
- **Analog Backend:** Smooths DAC output with 15.9 kHz filter, removes 1.65 V bias, sends signal to amplifier.

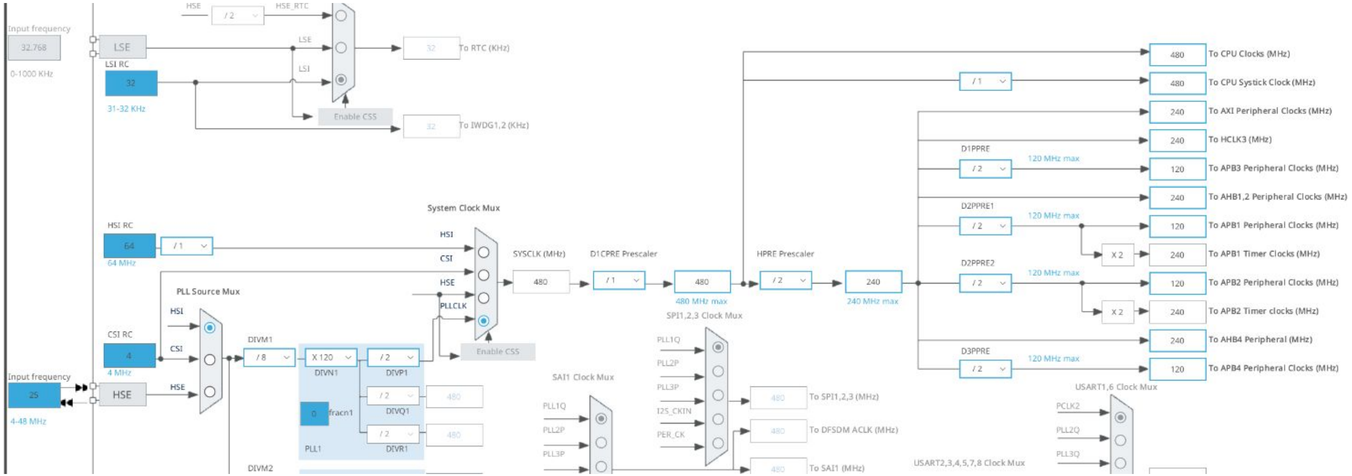
# Digital Hardware-to-Software Architecture and Concepts

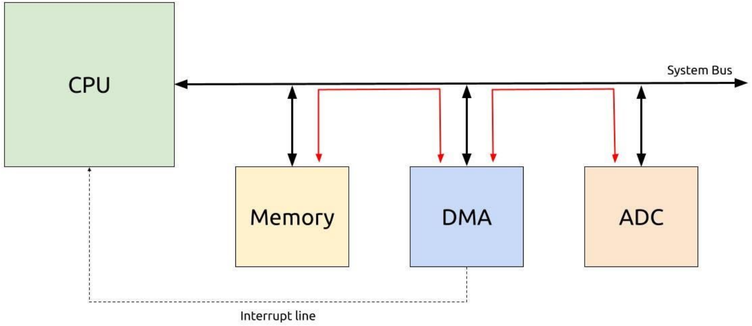
## How does the digital world interacts with the analog?

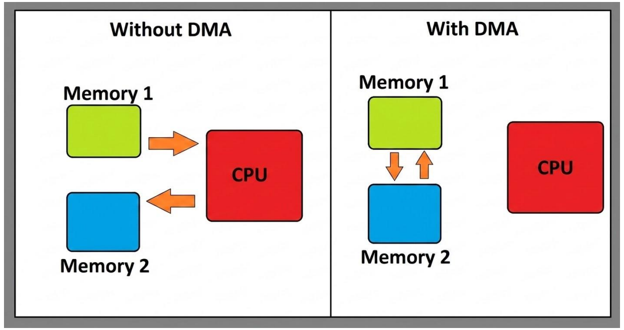



- **48 kHz Sample Rate:** TIM6 at 240 MHz / 5000 = 48 kHz. Triggers ADC and DAC each tick. CPU not involved.
- **ADC (16-bit):** Reads guitar voltage (0-3.3 V, biased at 1.65 V). Only upper 12 bits used to match DAC.
- **DMA:** Moves data between ADC/DAC and memory without CPU. CPU wakes only on half-buffer and full-buffer interrupts.
- **DAC (12-bit):** Converts processed samples to analog. 16-to-12 bit shift drops lower 4 bits, theoretical SNR floor = 74 dB.

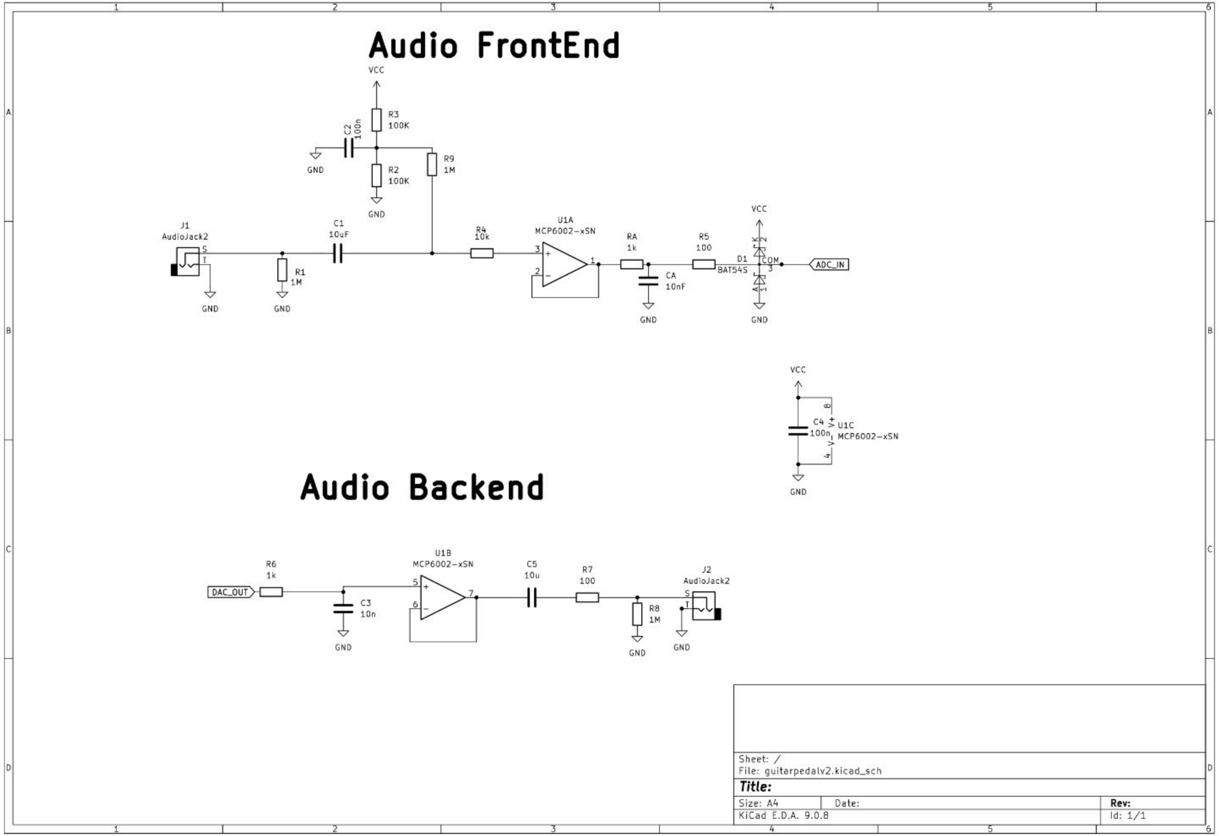
# DMA, Hardware Timers & Interrupts



# Analog Circuits Schematics




**What the circuit does?:**

- Match Impedance
- Center Voltage at 1.65V
- Buffer / Increase Signal Strength
- Filter Frequencies
- Protect the STM32 from Over Voltage or Current

Circuit Simulation

# DSP Echo Effect Algorithm

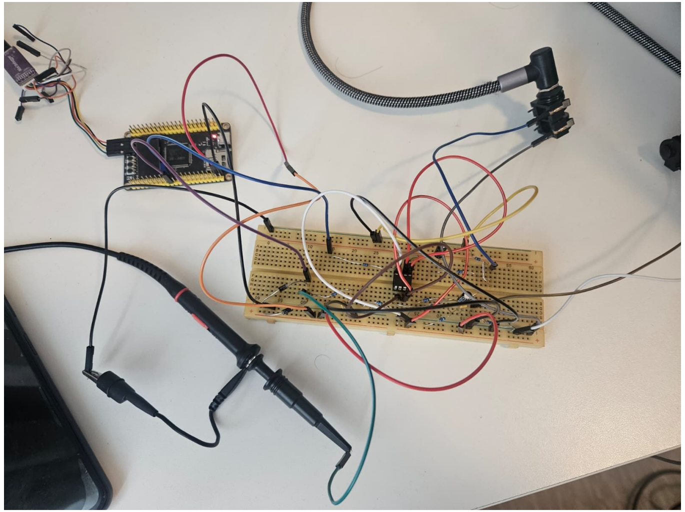
```c
#include "dsp.h"
#include <stdint.h>

__attribute__((section(
    ".dma_buffer"))) uint16_t delayBuffer[Delay];
uint16_t delayWriteIndex = 0;


void dsp_algorithm(uint16_t *input, uint16_t *output, int startindex,
                   int endindex) {

  const int32_t ADC_MID = 32768;
  const int32_t DAC_MID = 2048;

  for (int i = startindex; i < endindex; i++) {
    // Scale 16-bit ADC to 12-bit centered signal
    int32_t dry = ((int32_t)input[i] - ADC_MID) >> 4;

    // Read oldest sample from the delay line (Delay-1 samples ago)
    uint16_t readIndex = (delayWriteIndex + 1) % Delay;
    int32_t delayed = (int32_t)delayBuffer[readIndex] - DAC_MID;

    // Write dry + 50% feedback into the delay buffer
    int32_t fb = dry + (delayed / 2);
    if (fb >  2047) fb =  2047;
    if (fb < -2048) fb = -2048;
    delayBuffer[delayWriteIndex] = (uint16_t)(DAC_MID + fb);
    delayWriteIndex = (delayWriteIndex + 1) % Delay;

    // Output: 100% dry + 60% wet echo
    int32_t out = DAC_MID + dry + (delayed * 6 / 10);
    if (out > 4095) out = 4095;
    if (out < 0)    out = 0;

    output[i] = (uint16_t)out;
  }
}
```

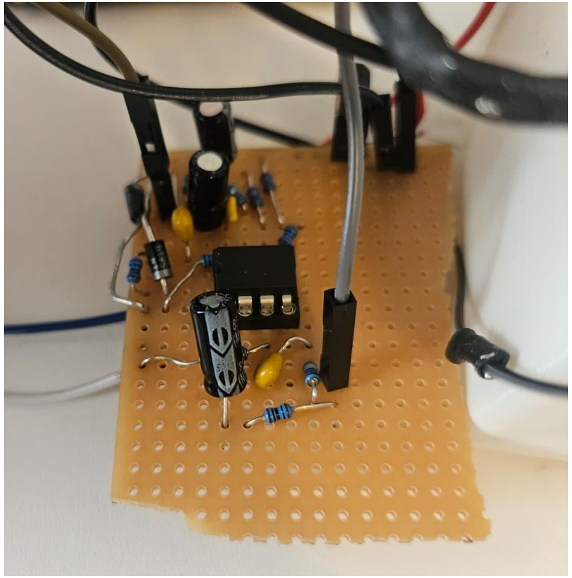
- Buffer D2 SRAM to be directly accessible to DMA (cannot reach DTCM)
- 14 400 samples at 48 kHz → about 300 ms of audio history
- Double Buffering with Half full DMA Callbacks. (Latency = 128 / 48 000 = 2.67 ms per half, 5.33 ms total)
- Only Index is changed, memory is not copied/re-allocated every iteration of circular-buffer

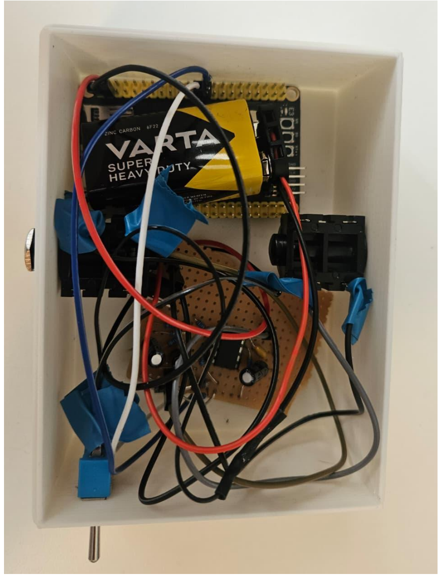
```c
/* USER CODE BEGIN 4 */
// second half of the buffer is filled
void HAL_ADC_ConvCpltCallback(ADC_HandleTypeDef *hadc) {

  dsp_algorithm(adcBuffer, dacBuffer, halfN, N);

}

// first half of the buffer is filled
void HAL_ADC_ConvHalfCpltCallback(ADC_HandleTypeDef *hadc) {
  dbg_halfCpltCount++;

  dsp_algorithm(adcBuffer, dacBuffer, 0, halfN);
}       You, 4 months ago • first commit: DMA, ADC, BUFFER are
/* USER CODE END 4 */
```

# Soldering and System Implementation of Electronics

- **Breadboard:** Each stage tested alone with signal generator before connecting to MCU.
- **Protoboard:** No ground plane, main source of noise (SNR = 20.83 dB).
- **Enclosure:** Holds protoboard + WeAct Mini board. 9 V battery through 3.3 V regulator (LM1117-3.3) all together.







# Development Tools Environment

- **STM32CubeMX:** Configures ADC, DAC, TIM6, DMA peripherals. Generates HAL init code.
- **arm-none-eabi-gcc v12 + CMake:** Cross-compiler for Cortex-M7. CMake generates Makefiles. (Provided by STM)
- **KiCad 8**
- **Falstad Circuit Simulator**
- **Audacity:** Audio recording and analysis of the output signal. (Freq Spectrum, Noise Calculations)
- **Oscilloscope**
- **Signal Generator**
- **Guitar Amp & Guitar**

# Conclusion

- Goal proposed was achieved. Echo effect is audible on real guitar input.
- Learned a lot about circuit design and how DMA and double buffering actually work in practice.
- Next steps: custom PCB with ground plane, Sallen-Key filters instead of single RC, true-bypass switch, mount STM32 directly on PCB to shrink size, add more effects like chorus/reverb/distortion with footswitch/BLE APP select.
- Audible noise at the output. Measured at 20.83 dB SNR, causes identified and justified (see report for more details).
- Functional Usable Prototype. Pedal works but is not a finished product. There is alot to build upon and refine

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